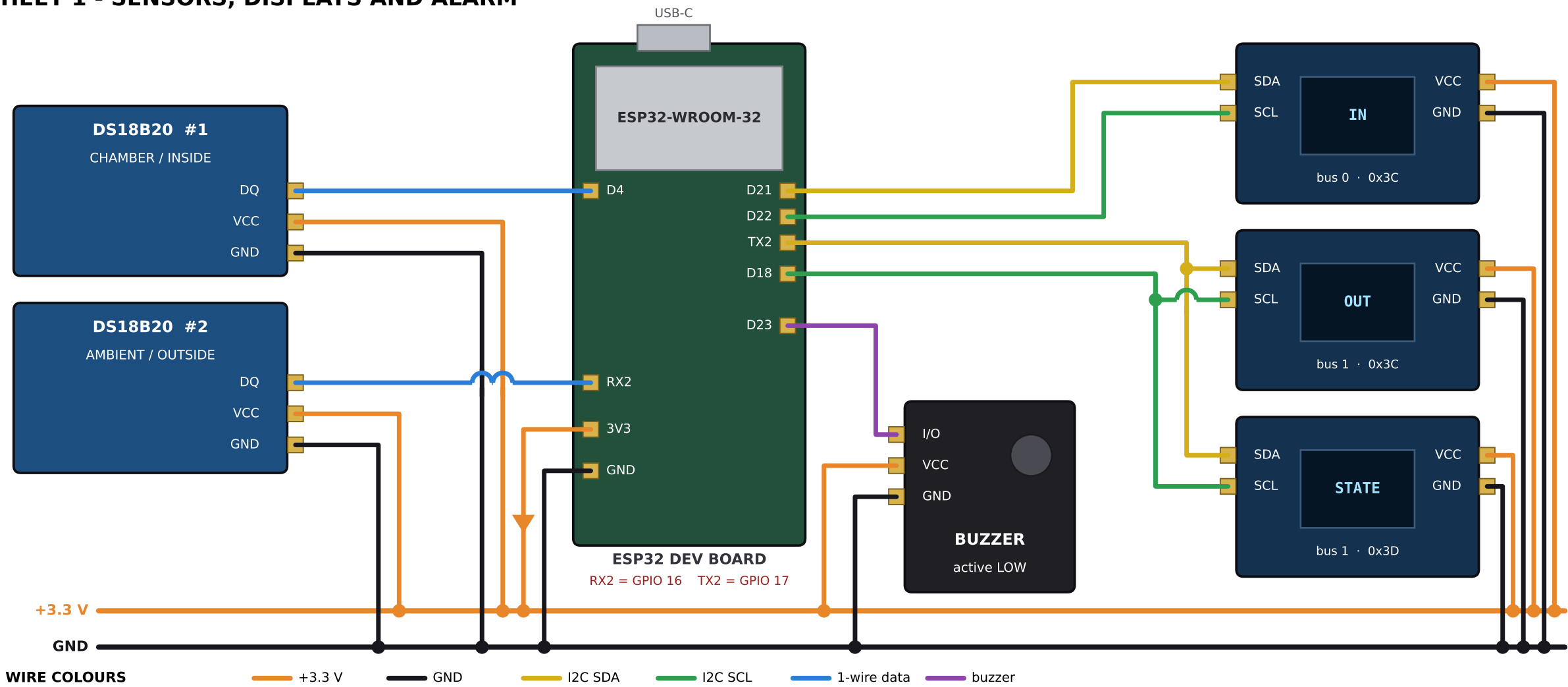


SHEET 1 - SENSORS, DISPLAYS AND ALARM



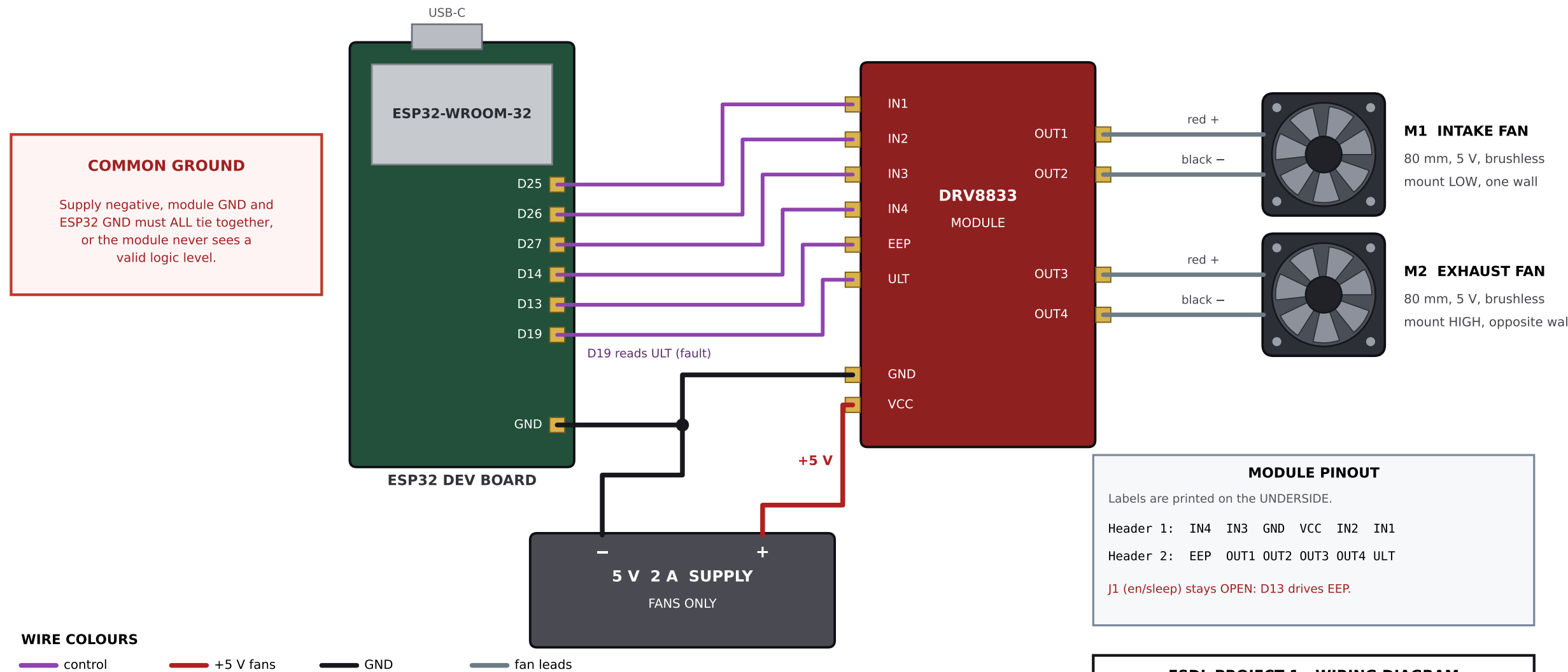
A wire that HOPS over another is NOT connected to it; a solid dot marks a real junction.
Pin headers are grouped for clarity - match pins by their printed LABEL, not by physical position.

3V3 is an OUTPUT of the dev board

USB-C powers the ESP32; its regulator feeds every part on this sheet. Never connect an external supply to 3V3.

ESDL PROJECT 1 - WIRING DIAGRAM		
Sensors, Displays and Alarm		
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SHEET 2 - FAN DRIVE AND POWER



A wire that HOPS over another is NOT connected to it; a solid dot marks a real junction.
Pin headers are grouped for clarity - match pins by their printed LABEL, not by physical position.

Fan power never passes through the ESP32, which runs from USB-C. Brushless fans: no suppression capacitors.

ESDL PROJECT 1 - WIRING DIAGRAM		
Fan Drive and Power		
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SHEET 3 - CONNECTION TABLE

Tick each row as you wire it. Build one component at a time and verify it with the serial console before the next.

	COMPONENT	PIN (as printed)	CONNECTS TO	WIRE	NOTE
☐	DS18B20 #1 (inside)	DQ	ESP32 D4 (GPIO 4)	blue	module has the 4.7k pull-up
☐		VCC / GND	+3.3 V rail / GND rail	orange / black	
☐	DS18B20 #2 (outside)	DQ	ESP32 RX2 (GPIO 16)	blue	printed RX2, not 16
☐		VCC / GND	+3.3 V rail / GND rail	orange / black	keep clear of exhaust plume
☐	OLED IN (bus 0, 0x3C)	SDA / SCL	ESP32 D21 / D22	yellow / green	
☐		VCC / GND	+3.3 V rail / GND rail	orange / black	3.3 V only - no shifter
☐	OLED OUT (bus 1, 0x3C)	SDA / SCL	ESP32 TX2 / D18	yellow / green	printed TX2 = GPIO 17
☐		VCC / GND	+3.3 V rail / GND rail	orange / black	
☐	OLED STATE (bus 1, 0x3D)	SDA / SCL	same bus 1 rows as OUT	yellow / green	address pad moved to 0x3D
☐		VCC / GND	+3.3 V rail / GND rail	orange / black	
☐	Buzzer module	I/O	ESP32 D23 (GPIO 23)	purple	active LOW: sounds when pulled low
☐		VCC / GND	+3.3 V rail / GND rail	orange / black	
☐	DRV8833 module	IN1 / IN2	ESP32 D25 / D26	purple	intake fan
☐		IN3 / IN4	ESP32 D27 / D14	purple	exhaust fan
☐		EEP	ESP32 D13 (GPIO 13)	purple	HIGH enables; leave J1 open
☐		ULT	ESP32 D19 (GPIO 19)	purple	fault output, open drain
☐		VCC	5 V fan supply +	red	NEVER from the ESP32
☐		GND	supply – AND ESP32 GND	black	COMMON GROUND - mandatory
☐		OUT1 / OUT2	intake fan red / black	grey	fan runs one direction only
☐		OUT3 / OUT4	exhaust fan red / black	grey	fan runs one direction only
☐	Power	ESP32 USB-C	computer USB port	-	powers the logic
☐		ESP32 3V3	+3.3 V rail	orange	an OUTPUT - sources the rail
☐		ESP32 GND	GND rail	black	
☐	Leave unwired	GPIO 33	nothing	-	bench-test reference pin

ESDL PROJECT 1 - WIRING DIAGRAM		
Connection Table		
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